

FARES GRINA, PhD

SENIOR DATA & AI ENGINEER

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PROFILE

Senior Data & AI Engineer with a PhD in artificial intelligence, specialising in generative models, uncertainty modelling and model evaluation. Builds AI agents, data pipelines and production applications, connecting research methods with practical engineering.

PROFESSIONAL EXPERIENCE

AI and R&D Engineer – Talan

Dec 2024–Present

Paris, France

Client engagement – Coface

Mar 2026–Present

Senior Data & AI Engineer / Business Information / Professional Services

Paris, France

- Developing an **MCP server and portfolio-analysis AI agent** that enables customers to query portfolio data in natural language, investigate credit-risk exposure, and support credit decisions.
- Re-engineered a **14-step manual bulk-order process into a 3-step workflow**, reducing turnaround from roughly **two hours to a few minutes**; implemented the solution with FastAPI, Airflow, React/Vite, Celery, and Redis.
- Designing and operating a **Snowflake data-engineering pipeline** to integrate Coface datasets into the Snowflake ecosystem and support distribution through Snowflake Marketplace.
- Designed a **client-specific API caching solution** to reduce redundant API calls, improve response times, and make API consumption more efficient.
- Building **API and order-monitoring dashboards** from application logs to analyse API usage, cancelled orders, processing failures, and recurring ordering issues.

Technologies: Python, FastMCP, AWS, Amazon Bedrock AgentCore, FastAPI, React/Vite, Airflow, Celery, Redis, SQL Server, PostgreSQL, Snowflake, Docker, GitLab CI/CD.

Client engagement – SNCF

Oct 2025–Jan 2026

AI Engineer / Generative-AI multi-agent framework

Saint-Denis, France

- Designed and developed a **multi-agent architecture** automating email handling, web research and navigation, audit-document generation, and memory management through manual, autonomous, and hybrid modes.
- Implemented **MCP servers, a persistent knowledge graph, and an automated browsing agent**; added unit, integration, and business tests and instrumented traces, costs, and evaluations.

Technologies: Python, Rust, TypeScript, MCP, LLM/VLM, REST, Playwright, knowledge graphs, LangFuse, GitHub.

Client engagement – Stade Français Rugby

Jan 2025–Aug 2025

AI/ML Engineer / EMG analysis and injury-risk modelling

Paris, France

- Processed raw electromyography signals and developed **time-series injury-risk models** to identify injury risk factors, validating results with the club's technical and medical staff.
- Built an **EMG analysis web application** backed by relational storage to automate visualisation, filtering, feature extraction, and annotation.

Technologies: Python, SciPy, scikit-learn, PyTorch, time-series Transformers, Flask, TypeScript, PostgreSQL.

R&D projects – Talan

2025–2026

Generative AI, LLM evaluation, and data products

Paris, France

- Proteus** – led a **five-person team** to propose a **novel methodology for evaluating LLM performance and robustness** on business tasks and implement it as a distributed platform for benchmarking, experiment tracking, dashboards, and model selection. Deployed the full multi-service platform on Azure, using REST and gRPC for service communication; co-authored and presented the paper at the **EvaLLM workshop, CORIA-TALN 2025**.
- RFP Analyzer:** built a RAG/MCP and vector-search workflow for traceable, compliance-checked tender responses.
- Other projects:** built time-series forecasting, SharePoint mission-assistant, and maritime-emissions pipelines.

INDEPENDENT EXPERIMENT

Jev versus classical text classifiers

Sep 2026

Built and published a reproducible Python benchmark on AG News, Banking77 and Emotion, comparing zero-shot Jev with TF-IDF classifiers and XGBoost. Analysed learning curves, calibration and paired-bootstrap uncertainty; released code and saved predictions.

github.com/faresGr/jev-classifier-benchmark

EDUCATION

PhD in Computer Science – Artificial Intelligence <i>University of Artois, France & University of Tunis (ISG), Tunisia</i> Thesis: <i>Dealing with Uncertainty in Imbalanced Classification using Belief Function Theory.</i> Research on uncertainty, evidential learning, ensemble methods and generative models.	2021–2024 <i>Arras, France</i>
Master’s in Computer Science for Decision Making <i>Institut Supérieur de Gestion de Tunis, University of Tunis</i> Machine learning, uncertainty modelling, optimisation and text mining.	2018–2020 <i>Tunis, Tunisia</i>
Bachelor’s in Business Informatics <i>Institut Supérieur de Gestion de Tunis, University of Tunis</i> Software engineering, databases, business intelligence and operations research.	2015–2017 <i>Tunis, Tunisia</i>

EARLIER EXPERIENCE

Teaching Assistant – University of Artois, IUT Béthune <i>Programming, networks and concurrent systems</i> Taught object-oriented programming, network programming, multithreading and Python.	Sep 2023–Dec 2024 <i>Béthune, France</i>
Research Intern – LARODEC <i>Uncertainty-aware machine learning</i> Proposed an uncertainty-aware oversampling method using belief function theory; evaluated it on noisy, overlapping and imbalanced classes.	Feb–Nov 2020 <i>Tunis, Tunisia</i>
Data Scientist – Epilert <i>Medical-signal machine learning and IoT</i> Built seizure-detection models from physiological signals and an IoT delivery stack with MQTT, ELK monitoring and CI/CD, in collaboration with the medical team.	Jul 2018–Feb 2019 <i>Tunis, Tunisia</i>
Cybersecurity Data Science Intern – Keystone Group <i>Network analytics and intrusion detection</i> Developed a NetFlow threat-detection pipeline, anomaly-detection models and a Django/ELK alerting interface with real-time correlation rules.	Feb–Jun 2018 <i>Tunis, Tunisia</i>

SELECTED PUBLICATIONS

- F. Grina and N. Kalashnikova, “**Évaluation de la Robustesse des LLM : Proposition d’un Cadre Méthodologique et Développement d’un Benchmark,**” EvalLLM workshop, CORIA-TALN, 2025.
- F. Grina, Z. Elouedi, and E. Lefèvre, “**Re-sampling of multi-class imbalanced data using belief function theory and ensemble learning,**” *International Journal of Approximate Reasoning*, vol. 156, pp. 1–15, 2023.
- F. Grina, Z. Elouedi, and E. Lefèvre, “**Evidential Generative Adversarial Networks for handling imbalanced learning,**” ECSQARU, 2023.
- Full publication list: faresgr.github.io/publications

TECHNICAL SKILLS

AI & ML Machine learning, deep learning, generative AI, LLMs, RAG, uncertainty modelling, evidential learning, imbalanced learning, model evaluation, explainable AI
Programming & data Python, SQL, JavaScript/TypeScript, PyTorch, TensorFlow, Keras, scikit-learn, pandas, NumPy
Cloud & AI platforms Microsoft Azure, Azure Machine Learning, AWS, Amazon Bedrock, Bedrock AgentCore
Backend & orchestration FastAPI, REST APIs, Apache Airflow, Celery, Redis, WireMock
Databases & platforms PostgreSQL, SQL Server, MongoDB, Snowflake, Elasticsearch
Applications & DevOps React, Vite, Streamlit, Docker, GitLab CI/CD, Git, Linux, SonarQube, Copilot Studio

CERTIFICATIONS & LANGUAGES

Certifications	Languages
Microsoft Azure AI Fundamentals (AI-900)	Arabic – Native
Professional Scrum Master I (PSM I)	French – Fluent
IBM Big Data Engineer – Mastery Award	English – Fluent
MTA Database Fundamentals	